

1. Let R be the finite region bounded by the graphs of $y = \ln(x)$, $y = 3$, $y = 5$, and $x = 0$. Set up a definite integral which represents the volumes of the following solids. Use your calculator to evaluate each integral.
 - (a) The volume of the solid with base R for which the cross-sections perpendicular to the y -axis are semi-circles.
 - (b) The volume of the solid formed when R is revolved around the y -axis. Determine this volume in the following two ways.
 - i. Integrate with respect to x .
 - ii. Integrate with respect to y .

2. Let R be the finite region bounded by the graphs of $y = \sqrt{x-3}$, $x = 7$, & $y = 0$. Set up a definite integral which represents the volumes of the following solids.
- (a) The volume of the solid with base R for which the cross-sections perpendicular to the x -axis are squares
- (b) The volume of the solid formed when R is revolved around the line $y = -1$. Determine this volume in the following two ways.
- Integrate with respect to x .
 - Integrate with respect to y .

3 – 4: Let R be the region bounded by the given curves in the first quadrant. Complete each of the following:

- (a) Sketch a picture of the region R and the solid generated when R is revolved about the given line.
- (b) Find the volume of the solid using the disk/washer method.
- (c) Find the volume of the solid using the shell method.
- (d) Verify that the results from (b) and (c) are the same.

3. $y = \sqrt{x}$, $y = x^3$ revolved around the x -axis.

4. $y = -x^2 + 5$, $x = 0$, & $y = 1$ revolved around the line $x = 3$.

6. Consider the region bounded by the curves $y = 3x$, $y = 12 - 3x$, & $y = 0$. If this region is revolved around the line $y = -1$ then answer the following questions.
- (a) Sketch a picture of the region.

- (b) What is the best method to compute the volume of the solid? Attempt to fill out the method in the table that you see as easier (if you are unsure observe both methods and then decide). Compute the volume.

Disks/Washers	Shells
Inner Radius (R):	Radius:
Outer Radius (r):	Height:
$\int_a^b [\pi R^2 - \pi r^2] d \text{---}$	$\int_a^b 2\pi(\text{radius})(\text{height}) d \text{---}$